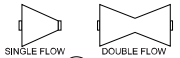
EQUIPMENT SYMBOLS

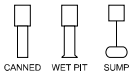


TURBINE

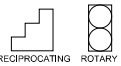

STEAM GENERATOR


REACTOR VESSEL

HORIZONTAL CENTRIFUGAL PUMP



VERTICAL PUMP

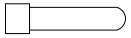


POSITIVE DISPLACEMENT PUMP


VACUUM PUMP


HAND PUMP

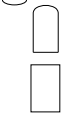

PLATE TYPE
HEAT EXCHANGER


U-TUBE HEAT EXCHANGER



STRAIGHT TUBE
HEAT EXCHANGER


PRESSURE VESSEL, VERTICAL
OR HORIZONTAL TANK


DOME ROOF TYPE TANK

CYLINDRICAL TYPE TANK
(DRUM TYPE)


CONE ROOF TANK

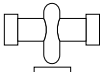

D : DIAPHRAGM TANK (AS SHOWN)
F : FLOATING TANK COVER


FLAT TYPE TANK


CONE BOTTOM TYPE TANK
OR CYCLONE



AIR COMPRESSOR
RE : RECIPROCATING (SHOWN)
RO : ROTARY
SC : SCREW
CF : CENTRIFUGAL


AFTER COOLER


GENERIC BOX EQUIPMENT


GENERIC CIRCLE EQUIPMENT


HOPPER

MIXER


ELECTRIC MOTOR


FUME SCRUBBER


PRESSURIZED GAS BOTTLE


BALL STRAINER


CHUTE


TANDEM LOUVER DAMPER


AMMONIA VAPORIZER


SWING PANEL


IRWST SUMP STRAINER

VALVE SYMBOLS

LINE VALVE-BODIES

OPEN DURING NORMAL OPERATION	CLOSED DURING NORMAL OPERATION	VALVE TYPE
V001A (VALVE NO.)		GATE
		GLOBE (ARROW INDICATES VALVE BODY ORIENTATION. THIS ARROW SHOW ONLY WHEN DIFFERENT FROM NORMAL SYSTEM FLOW DIRECTION OR REQUIRED IN BI-DIRECTION FLOW.)
NO	NC	BUTTERFLY
		BALL (2-WAY)
		BALL VALVE (3-WAY)
		PLUG (2-WAY)
		PLUG VALVE (3-WAY)
		PLUG VALVE (4-WAY)
		ANGLE
		ANGLE CHECK VALVE
		DIAPHRAGM
BW	BW	PACKLESS VALVE B : BELLOWS SEAL (AS SHOWN) M : METALLIC DIAPHRAGM
		NEEDLE
		FOOT VALVE
		NON-RETURN VALVE
		RAM VALVE
		BACKFLOW PREVENTER
		CHECK (IDENTIFY CONTROLLED CLOSURE CHECK VALVES AND TILTING DISC CHECK VALVE BY NOTE)
NO	NC	STOP-CHECK
		SINGLE END SHUTOFF WITH QUICK DISCONNECT
		FLOW LIMITING VALVE

KNIFE GATE VALVE

DOUBLE BLADE DAMPER

VALVE POSITION STATUS (TYPICAL FOR ALL VALVES)



LO : LOCKED OPEN IN LOCAL



LC : LOCKED CLOSED IN LOCAL



LO: LOCKED OPEN ON MCB AND/OR REMOTE SHUTDOWN CONSOLE

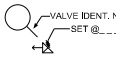


LC: LOCKED CLOSED ON MCB AND/OR REMOTE SHUTDOWN CONSOLE

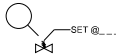


LT : LOCKED THROTTLED

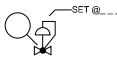
SELF-ACTUATED VALVES



ANGLE SAFETY
OR RELIEF



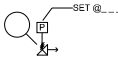
STRAIGHT SAFETY
OR RELIEF



PRESSURE REDUCING REGULATOR
SELF-CONTAINED



VACUUM RELIEF



PILOT OPERATED
SAFETY RELIEF

OPERATOR SYMBOLS
(FOR USE WITH VALVE & DAMPER SYMBOL)

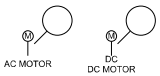
VALVE IDENTIFICATION



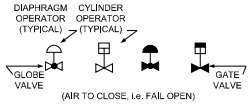
INSTRUMENT IDENTIFICATION NUMBER FOR CONTROL AIR & INSTRUMENT AIR SIGNAL SOLENOID VALVES ONLY


VALVE IDENTIFICATION NUMBER (TYPICAL FOR POWER OPERATED VALVES EXCEPT AS NOTED FOR CERTAIN SOLENOID VALVES)

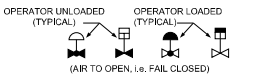
MOTOR OPERATOR



PNEUMATIC OPERATORS (SPRING RETURN)
FOR CONTROL AND ON/OFF VALVES
(TYPICAL FOR OTHER VALVE TYPES)



(AIR TO CLOSE, i.e. FAIL OPEN)



(AIR TO OPEN, i.e. FAIL CLOSED)

SOLENOID OPERATORS
(SPRING RETURN) FOR VALVES
(TYPICAL FOR OTHER VALVE TYPES)



OTHER OPERATORS

DOUBLE ACTING
(AIR TO OPEN AND AIR TO CLOSE)


ELECTRO-HYDRAULIC ACTUATORS

HAND ACTUATOR
(NOT SHOWN UNLESS AS INDICATED ABOVE OR WHEN USED AS AN EXTENSION SHAFT THRU SHIELD WALL OR FOR FLOORSTAND)

MANUAL FLOW CONTROL

OPERATOR FAILURE MODE ABBREVIATION



FO : FAIL OPEN



FC : FAIL CLOSED



FL : FAIL LOCKED IN PLACE
(POSITION DOES NOT CHANGE)

MISCELLANEOUS PIPING SPECIALTY SYMBOLS



TRAP
ST : STEAM TRAP (SHOWN)
AT : AIR TRAP
DT : DRAIN TRAP
RT : RESIN TRAP


Y-TYPE STRAINER

TEMPORARY STARTUP STRAINER


CIRCULAR OR HAMMER BLIND


SPECTACLE BLIND


SIMPLEX BASKET STRAINER


DUPLEX BASKET STRAINER


MIXING TEE

BURSTING PLUG

RUPTURE DISC FOR PRESSURE RELIEF

RUPTURE DISC FOR VACUUM RELIEF

RESTRICTION ORIFICE
(SINGLE OR MULTI-STAGE)



RESTRICTION DEVICE
(CAPILLARY)


CAVITATING VENTURI

EXPANSION JOINT

FLEXIBLE HOSE


FLEXIBLE CONNECTION


PULSATION DAMPENER



FLAME ARRESTOR


OIL SEAL POT



FLOW SIGHT GLASS


FLOW STRAIGHTENING VANE



FILTER



DESICCANT FILTER


ROOF EXHAUST HOOD


SPRAY NOZZLE OR SPARGER
(UPWARD OR DOWNWARD)


ELECTRIC HEATER

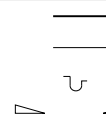

AIR SEPARATOR ASSEMBLY

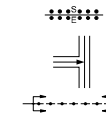

CONDENSATE POT



PIPE LIN







MISC



HVAC EQUIPMENT / DAMPERS SYMBOLS


CENTRIFUGAL FAN OR BLOWER


VANE AXIAL FAN


HUMIDIFIER



HEATING COIL(HC)
EL : ELECTRIC (SHOWN)
HW : HOT WATER
S : STEAM



COOLING COIL (CC)
CW : CHILLED WATER (SHOWN)
CCW : COMPONENT COOLING WATER
DX : DIRECT EXPANSION


PREFILTER


HIGH EFFICIENCY PARTICULATE AIR
(HEPA) FILTER



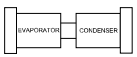
CARBON ADSORBER


POSTFILTER


MOISTURE SEPARATOR


ROOF SUPPLY FAN


ROOF VENTILATOR
WITH PARALLEL BLADE DAMPER



CHILLER



CUBICLE COOLER
WITH CENTRIFUGAL FAN



NORMALLY OPEN DAMPER
(PNEUMATIC OPERATED)



NORMALLY CLOSED DAMPER
(PNEUMATIC OPERATED)



NORMALLY OPEN DAMPER
(ELECTRO-HYDRAULIC OPERATED)

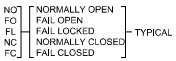


NORMALLY CLOSED DAMPER

HVAC EQUIPMENT / DAMPERS SYMBOLS

DUCTED AIR FLOW


NON-DUCTED AIR FLOW





SMOKE DAMPER

TEST CONNECTIONS



PRESSURE TEST CONNECTION



GRAB SAMPLE TEST CONECTION



TEMPERATURE TEST CONNECTION
(WHEN THERMOWELL IS NOT USED)



FLOW TEST CONNECTION



SPECIAL TEST CONNECTION
(DESCRIBE)



SPECIAL TEST CONNECTION
(DESCRIBE)


ASME TEST POINT ONLY(TYP.)


BOTH ASME AND NORMAL TEST POINT
(TYP.)

* REFER TO ABOVE TEST PARAMETER :
P, G, T, etc. .

ABBREVIATIONS

HVAC ONLY

OA : OUTSIDE AIR
ATM : ATMOSPHERE
AHU : AIR HANDLING UNIT
ACU : AIR CLEANING UNIT
CC : CUBICLE COOLER
PACU : PACKAGED AIR CONDITIONING UNIT
SR : SAFETY RELATED
NSR : NON-SAFETY RELATED

FIRE PROTECTION SYMBOLS *


OUTDOOR FIRE HYDRANT


OUTDOOR FIRE HYDRANT WITH
HOSE HOUSE


ANGLE VALVE OF INDOOR
FIRE HYDRANT


ALARM CHECK VALVE


POST INDICATOR VALVE


HYDRANT ISOLATION GATE VALVE


DRY VALVE


DRY VALVE WITH
QUICK-OPENING DEVICE
(ACCELERATOR OR EXHAUSTER)


CLEAN AGENT, CO_2, OR GAS
SPRAY NOZZLE


DELUGE VALVE


SPRINKLER OR WATER
SPRAY NOZZLE

AUTO DRIP VALVE


INDOOR FIRE HYDRANT WITH HOSE
HOUSE (WET TYPE)


INDOOR FIRE HYDRANT WITH HOSE
HOUSE (DRY TYPE)


CO_2 HOSE REEL STATION


FOAM HOSE STATION


FOAM CHAMBER


FOAM PROPOTIONER

* NOTE : FOR SPRAY NOZZLES OR
SPRINKLER HEADS. USE THE
PIPING SPECIALLTY SYMBOL
FOR SPRAY NOZZLE


SIAMESE FIRE
HOUSE CONNECTION


SUPPRESSION GAS BOTTLE


MANUAL PULL STATION ALARM



TEST BOX


SPEAKER / HORN


BELL / GONG



FIRE SYSTEM
WATER MOTOR ALARM

CARBON ADSORBER DELUGE SYSTEMS
NOZZLES (HVAC)

ABBREVIATIONS



FLOW NOZZLE









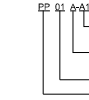
SINGLE PORT











* OPTI

AC = ACCI
AP = AIR F
BC = BATT
CD = CON
CH = CHIL
CM = COM
DD = DEM
DW = DEH
EC = ECOI
FT = FILTE
HE = HEA
PA = I&C E

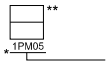
INSTRUMENT & CONTROL DEVICE SYMBOLS



SHARED HUMAN-SYSTEM INTERFACE (HSI)
(HAND SWITCH, INDICATOR, CONTROL, ALARM, RECORDER)
WITHIN POWER BLOCK (1)

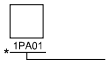


SHARED HSI (HAND SWITCH, INDICATOR,CONTROL, ALARM, RECORDER)
OUT OF POWER BLOCK (1)



MAIN CONTROL ROOM PANEL MOUNTED DEVICE (FIXED)

PANEL NUMBER (TYPICAL)



CONTROL DEVICE AND / OR CCS CABINET MOUNTED DEVICE
INSTALLED WITHIN POWER BLOCK

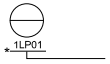
PANEL NUMBER (TYPICAL)



CONTROL DEVICE AND / OR CCS CABINET MOUNTED DEVICE
INSTALLED OUT OF POWER BLOCK



LOCALLY MOUNTED DEVICE NOT ON
INSTRUMENT PANEL OR ASSEMBLY



LOCAL INSTRUMENT PANEL
MOUNTED DEVICE

PANEL NUMBER (TYPICAL)



MULTIPLE DEVICES IN ONE ASSEMBLY
(i.e. TEMPERATURE ELEMENT IN THERMOWELL)

* PLANT UNIT NUMBER NOT REQUIRED IF SAME AS DRAWING PLANT UNIT NUMBER

** THE INDICATED POWER SEGREGATION REPRESENTS THE POWER SEGREGATION OF COMPONENT BEING CONTROLLED BY THE SHARED CONTROL.

*** TYPE OF HSI DEVICE SUCH AS QIAS-N, QIAS-P OR LDP IS PRESENTED AS SUFFIX.
HOWEVER, THE TYPE OF HSI DEVICE IS NOT REQUIRED IF THE SYMBOL IS USED FOR THE SHARED CONTROL.

THE SHARED SYMBOL WITHOUT SUFFIX MEANS THIS PARAMETER IS ONLY REPRESENTED ON THE INFORMATION FPD. THE QIAS-N MEANS THIS PARAMETER IS REPRESENTED ON BOTH INFORMATION FPD AND QIAS-N FPD.
THE LDP MEANS THIS PARAMETER IS REPRESENTED ON THE INFORMATION FPD, QIAS-N AND LDP.
FOR A MAN SWITCH, ONLY THE COMPONENT STATUS IS DISPLAYED ON QIAS-N AND LDP.

(1) SWITCH YARD IS EXCLUDED FROM POWER BLOCK AREA.

COMPUTER INPUT - OUTPUT SIGNAL NUMBER

ALL OF INPUT POINTS TO CCS ARE REGARED AS THE COMPUTER INPUT DATA FOR IPS OR QIAS-N.

LOCAL PANEL INPUT SYMBOL



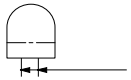
INPUT SIGNAL FROM LOCAL PANEL TO COMPUTER THROUGH CCS



INPUT SIGNAL TO QIAS-P COMPUTER

CONTROL & INSTRUMENTATION FUNCTION SYMBOLS & CODES

FUNCTION SYMBOLS

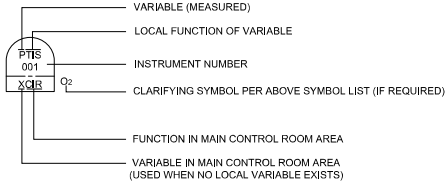


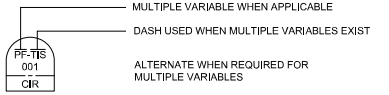
ELECTRICAL SIGNAL BETWEEN LOCAL & MAIN CONTROL ROOM / I & C EQUIPMENT ROOM FUNCTIONS
CONNECTION FOR THE PROCESS VARIABLE (TYPICAL)



LOCAL FUNCTION

FUNCTION SYMBOL CODES





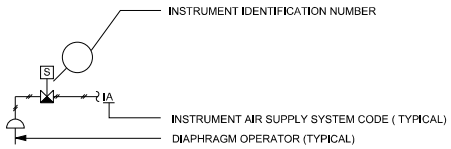
INSTRUMENT TYPE CODES

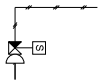
FIRST LETTER VARIABLE (MEASURED OR INITIATING)	
A	ANALYSIS
B	BURNER
C	CONDUCTIVITY
D	DENSITY
E	VOLTAGE
F	FLOW
H	HAND (MANUAL)
I	CURRENT (ELEC.)
J	POWER (ELEC.)
K	TIME
L	LEVEL(LIQUID OR SOLID)
M	MOISTURE (HUMIDITY)
N	NEUTRON FLUX
P	PRESSURE OR VACUUM
PD	DIFFERENTIAL PRESSURE
R	RADIOACTIVITY
S	SPEED OR FREQUENCY
T	TEMPERATURE
U	MULTIVARIABLE
V	VIBRATION
W	WEIGHT OR FORCE
X	SPECIAL (REFER TO CLARIFYING SYMBOL LIST)
Y	STRAIN
Z	POSITION

SECOND OR SUCCEEDING LETTERS INSTRUMENT FUNCTION	
A	ALARM
AC	ALARM CLOSED
AH	ALARM HIGH
AHH	ALARM HIGH HIGH
AHL	ALARM HIGH OR LOW
AL	ALARM LOW
ALL	ALARM LOW LOW
AO	ALARM OPEN
C	CONTROLLER (BLIND)
E	PRIMARY ELEMENT
EW	PRIMARY ELEMENT WELL
G	GLASS
I	INDICATOR
IC	INDICATING CONTROLLER
IK	INDICATING CONTROL STATION (MANUAL/AUTO)
IR	INDICATING RECORDER
IS	INDICATING SWITCH
IT	INDICATING TRANSMITTER
IY	INDICATING RELAY
JR	MULTIPOINT RECORDER (MORE THAN 3 PENS)
K	CONTROL STATION-VARIABLE TYPE (MANUAL/AUTO) WITHOUT CONTROLLER
L	LIGHT (PILOT)
Q	INTEGRATOR
R	RECORDER
RC	RECORDER CONTROLLER
S	SWITCH
T	TRANSMITTER

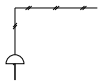
TYPICAL INSTRUMENT & CONTROL SYMBOL EXAMPLES

MULTIPLE CONTROL SIGNALS FOR PNEUMATIC OPERATORS

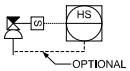




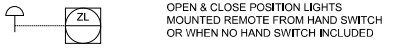
SINGLE CONTROL SIGNALS FOR PNEUMATIC OPERATORS



VALVE / DAMPER POSITION SIGNALS



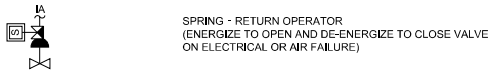




VALNE OPERATOR CONTROLS
(TYPICAL FOR OTHER CONTROL DEVICES)







INSTRUMENT & CONTROL SIGNAL SYMBOLS

